A Konyagin-Type Lower Bound in an Erdős Divisibility Problem

1 Introduction

For a positive integer k, let n_k be the least integer $n > 2k$ such that

$$p \nmid \prod_{1 \le i \le k} (n - i)$$

for every prime p with $k < p < 2k$. This is Erdős Problem #451 in the Erdős Problems website [2]. Equivalently, n_k is the least integer $n > 2k$ for which no prime in $(k, 2k)$ divides any of the k consecutive integers

$$n - k,\ n - k + 1, \dots,\ n - 1.$$

Such integers exist. Indeed, if

$$Q_k = \prod_{k < p < 2k} p,$$

then every sufficiently large multiple n of Q_k satisfies

$$n - i \equiv -i \not\equiv 0 \pmod p$$

for all $1 \le i \le k$ and all primes $p \in (k, 2k)$.

We shall call an integer $n > 2k$ *k-admissible* if it satisfies the above divisibility condition. Thus n_k is the least k-admissible integer.

The purpose of this note is to prove the lower bound

$$n_k > \exp\left(c \frac{(\log k)^2}{\log \log k} \right)$$

with an absolute constant $c > 0$.

The idea is simple. We use only primes very close to k. Let

$$H = k^{1-\beta},$$

where $0 < \beta < 1$ is fixed. If $p \in (k, k + H]$ and n is k-admissible, then the least nonnegative residue of n modulo p is either 0, or larger than k. Since $p - k \le H$, this implies

$$\left\| \frac{n}{p} \right\| \le \frac{H}{k} = k^{-\beta},$$

where $\|x\|$ denotes the distance from x to the nearest integer. Thus every prime $p \in (k, k + H]$ gives an integer point $u = p$ near the curve

$$v = \frac{n}{u}.$$

A theorem of Konyagin on integer points close to a smooth curve gives a strong upper bound for the number of such u, while the Baker–Harman theorem on primes in short intervals gives a lower bound of order $H/\log k$. With a careful choice of parameters these bounds are incompatible when

$$n \le \exp\left(c\frac{(\log k)^2}{\log\log k}\right).$$

2 Main result

Theorem 1. *There exists an absolute constant $c > 0$ such that, for all sufficiently large k,*

$$n_k > \exp\left(c\frac{(\log k)^2}{\log\log k}\right).$$

Equivalently, for all sufficiently large k, every integer n satisfying

$$2k < n \le \exp\left(c\frac{(\log k)^2}{\log\log k}\right)$$

has the property that

$$\prod_{1\le i\le k}(n-i)$$

is divisible by at least one prime $p \in (k, 2k)$.

We first record a preliminary lower bound. The proof is included to make the argument self-contained.

Lemma 1. *There exists an absolute constant $c_1 > 0$ such that, for all sufficiently large k, every k-admissible integer n satisfies*

$$n \ge c_1\frac{k^2}{\log k}.$$

Proof. We use the prime number theorem with the classical de la Vallée Poussin error term; see, for example, [3, Chapter 15]. In particular, uniformly for $k \le u < v \le 2k$,

$$\pi(v) - \pi(u) = \int_u^v \frac{\mathrm{d}t}{\log t} + O\Big(k\exp(-c\sqrt{\log k})\Big)$$

for some absolute constant $c > 0$. Hence every interval $I \subset (k, 2k)$ of length $|I| \ge k/(\log k)^2$ contains a prime, once k is sufficiently large.

Let n be k-admissible and put

$$T = \frac{n}{k}.$$

We first prove that, for every fixed $C > 0$, no k-admissible n can satisfy

$$T \le C\log k$$

for all sufficiently large k.

We need a simple geometric observation. For every real $T > 2$, there is an integer $a \ge 1$ such that

$$\left(\frac{T-1}{a}, \frac{T}{a}\right) \cap (1, 2)$$

has length at least $1/(4T)$. Indeed, write

$$T = m + \alpha, \qquad m \geq 2, \quad 0 \leq \alpha < 1.$$

If $\alpha \geq 1/2$, take $a = m$. Then

$$\left(\frac{T-1}{m}, \frac{T}{m}\right) \cap (1,2) = \left(1, 1 + \frac{\alpha}{m}\right),$$

whose length is at least $1/(4T)$. If $\alpha < 1/2$ and $m = 2$, take $a = 1$; the intersection is $(1+\alpha, 2)$, whose length is larger than $1/2$. Finally, if $\alpha < 1/2$ and $m \geq 3$, take $a = m - 1$. Then

$$\left(\frac{T-1}{m-1}, \frac{T}{m-1}\right) = \left(1 + \frac{\alpha}{m-1}, 1 + \frac{1+\alpha}{m-1}\right) \subset (1,2),$$

and its length is $1/(m-1) \geq 1/(4T)$.

Now suppose $T \leq C \log k$. Choose a as above. Multiplying by k, the interval

$$\left(\frac{n-k}{a}, \frac{n}{a}\right) \cap (k, 2k)$$

has length at least

$$\frac{k}{4T} \geq \frac{k}{4C \log k}.$$

For sufficiently large k, this length is larger than $k/(\log k)^2$, so the interval contains a prime $p \in (k, 2k)$. Thus

$$\frac{n-k}{a} < p < \frac{n}{a},$$

and therefore

$$n - k < ap < n.$$

Let

$$i = n - ap.$$

Then i is an integer satisfying $0 < i < k$, so $1 \leq i \leq k-1$, and

$$p \mid n - i.$$

This contradicts k-admissibility. Therefore, for every fixed $C > 0$, every k-admissible n satisfies

$$T > C \log k$$

for all sufficiently large k.

We now strengthen this to $T \gg k/\log k$. Put

$$L_0 = (\log k)^2, \qquad H_0 = \left\lfloor \frac{k}{L_0} \right\rfloor,$$

and let

$$\mathcal{P} = \{p \text{ prime} : k < p \leq k + H_0\}.$$

By the prime number theorem,

$$|\mathcal{P}| \geq c_0 \frac{k}{L_0 \log k}$$

for some absolute constant $c_0 > 0$ and all sufficiently large k. Since $H_0 < k$, we have

$$\mathcal{P} \subset (k, 2k).$$

Let $p \in \mathcal{P}$. Since n is k-admissible, the least nonnegative residue r_p of n modulo p satisfies either $r_p = 0$, or $k+1 \le r_p \le p-1$. Hence there is an integer j_p such that

$$0 \le j_p \le p - k - 1 \le H_0 - 1$$

and

$$p \mid n + j_p.$$

Define

$$a_p = \frac{n + j_p}{p}.$$

Then

$$n \le a_p p < n + H_0.$$

Let

$$\mathcal{A} = \left\{ a \in \mathbb{Z} : \frac{n}{k + H_0} \le a < \frac{n + H_0}{k} \right\}.$$

Every a_p lies in $\mathcal{A}$. Writing $h = H_0/k$, we have $h \le 1/L_0$, and the length of the interval defining $\mathcal{A}$ is

$$\frac{n + H_0}{k} - \frac{n}{k + H_0} = T + h - \frac{T}{1+h} = \frac{Th}{1+h} + h \le \frac{T+1}{L_0}.$$

Thus

$$|\mathcal{A}| \le \frac{T}{L_0} + 2.$$

For a fixed $a \in \mathcal{A}$, the primes $p \in \mathcal{P}$ with $a_p = a$ satisfy

$$n \le ap < n + H_0.$$

Hence they lie in an interval of length H_0/a. Since

$$a \ge \frac{n}{k + H_0} = \frac{T}{1 + H_0/k} \ge \frac{T}{2}$$

for large k, the number of such primes is at most

$$1 + \frac{H_0}{a} \le 1 + \frac{2H_0}{T}.$$

Therefore

$$|\mathcal{P}| \le \left(\frac{T}{L_0} + 2\right)\left(1 + \frac{2H_0}{T}\right).$$

Using $H_0 \le k/L_0$, this gives

$$|\mathcal{P}| \le \frac{T}{L_0} + 2 + \frac{2k}{L_0^2} + \frac{4k}{L_0 T}.$$

Combining this with the lower bound for $|\mathcal{P}|$, and multiplying by L_0, we obtain

$$c_0 \frac{k}{\log k} \le T + 2L_0 + \frac{2k}{L_0} + \frac{4k}{T}.$$

Now take $C = 16/c_0$ in the first part of the proof. For all sufficiently large k,

$$T > C \log k.$$

Thus

$$\frac{4k}{T} \le \frac{c_0}{4} \frac{k}{\log k}.$$

Also,

$$2L_0 + \frac{2k}{L_0} = o\left(\frac{k}{\log k}\right).$$

Hence, for sufficiently large k,

$$2L_0 + \frac{2k}{L_0} \le \frac{c_0}{4} \frac{k}{\log k}.$$

It follows that

$$c_0 \frac{k}{\log k} \le T + \frac{c_0}{2} \frac{k}{\log k},$$

and so

$$T \ge \frac{c_0}{2} \frac{k}{\log k}.$$

Since $n = Tk$, the lemma follows with $c_1 = c_0/2$. $\square$

We shall also use primes in short intervals.

Lemma 2 (Baker–Harman)**.** *There is an absolute constant $\alpha > 0$ such that the following holds. If $0 < \beta < \alpha$, then, for all sufficiently large x,*

$$\#\{p \text{ prime} : x < p \le x + x^{1-\beta}\} \gg_\beta \frac{x^{1-\beta}}{\log x}.$$

One may take any fixed $\beta < 0.465$.

Proof. We use the Baker–Harman theorem [1] in the following standard form: there is an admissible constant $\alpha > 0$, with $\alpha = 0.465$ admissible, such that uniformly for sufficiently large x and

$$x^{1-\alpha} \le y \le x,$$

one has

$$\pi(x+y) - \pi(x) \gg \frac{y}{\log x}.$$

Taking $y = x^{1-\beta}$, with any fixed $0 < \beta < \alpha$, gives the stated estimate. $\square$

The main analytic input is Konyagin's integer-point theorem. We state precisely the form needed below.

Lemma 3 (Konyagin's integer-point theorem)**.** *Let $r > 1$ be an integer, $N > 0$, $W > 1$, and let $f \in C^{r+1}[0, N]$. Let $D_r > 0$, $D_{r+1} > 0$, $\delta > 0$, and $\lambda \ge 1$. Suppose that, for all $u \in [0, N]$,*

$$\frac{D_r}{2} \le f^{(r)}(u) \le D_r, \qquad |f^{(r+1)}(u)| \le D_{r+1}.$$

Let $S \subset [0, N] \cap \mathbb{Z}$, and suppose that for every $u \in S$ there exist $v \in \mathbb{Z}$ and $w \in \mathbb{N}$, with $w \le W$, such that

$$\left| f(u) - \frac{v}{w} \right| < \delta.$$

Then

$$|S| \le CN\left[(D_r\lambda^r W^2)^{1/(2r-1)} + \left(\frac{\delta W^2}{D_r\lambda^r}\right)^{1/(r-1)} + \left(\frac{D_{r+1}\lambda W^2}{D_r}\right)^{1/(2r)}\right] + 2r\lambda,$$

where $C > 0$ is an absolute constant.

Proof. This is Theorem 2, formula (4), of Konyagin [4]. Konyagin states throughout that all constants denoted by c_i are absolute; in particular the constant in this estimate is independent of $r, N, W, D_r, D_{r+1}, \delta$, and λ. □

We shall use only the following specialization to the curve A/u.

Lemma 4 (Konyagin's estimate for A/u)**.** *Let M, L be positive integers with $M \ge 2$ and $1 \le L \le M$. Let $A > M$, let $0 < \Delta < 1/2$, and let $r \ge 2$ be an integer satisfying*

$$\frac{(r+1)L}{M} \le \log 2.$$

Put

$$D = \frac{Ar!}{M^{r+1}}.$$

For

$$\mathcal{R}(A; M, L, \Delta) := \left\{u \in (M, M+L] \cap \mathbb{Z} : \left\|\frac{A}{u}\right\| < \Delta\right\},$$

one has, for every $\lambda \ge 1$,

$$|\mathcal{R}(A; M, L, \Delta)| \le CL\left[(D\lambda^r)^{1/(2r-1)} + \left(\frac{\Delta}{D\lambda^r}\right)^{1/(r-1)} + \left(\frac{(r+1)\lambda}{M}\right)^{1/(2r)}\right] + 2r\lambda,$$

where $C > 0$ is an absolute constant.

Proof. Apply Lemma 3 on the interval $[0, L]$ to

$$f(t) = (-1)^r \frac{A}{M+t}.$$

Then

$$f^{(r)}(t) = \frac{Ar!}{(M+t)^{r+1}} > 0.$$

Since $(r+1)L/M \le \log 2$, we have, for $0 \le t \le L$,

$$\frac{D}{2} \le \frac{Ar!}{(M+t)^{r+1}} \le D.$$

Moreover,

$$|f^{(r+1)}(t)| = \frac{A(r+1)!}{(M+t)^{r+2}} \le \frac{A(r+1)!}{M^{r+2}} = \frac{(r+1)D}{M}.$$

Now let

$$S = \{t \in [0, L] \cap \mathbb{Z} : M + t \in \mathcal{R}(A; M, L, \Delta)\}.$$

Since M is an integer, the map $t \mapsto M + t$ is a bijection from S onto $\mathcal{R}(A; M, L, \Delta)$. Hence

$$|S| = |\mathcal{R}(A; M, L, \Delta)|.$$

For every $t \in S$, there exists an integer $m \in \mathbb{Z}$ such that

$$\left| \frac{A}{M+t} - m \right| < \Delta.$$

Hence

$$|f(t) - (-1)^r m| < \Delta.$$

Thus the rational-approximation hypothesis of Lemma 3 is satisfied with

$$v = (-1)^r m, \qquad w = 1.$$

Since Lemma 3 is stated for $W > 1$, we may simply take $W = 2$; the resulting powers of W are absolute constants and will be absorbed into the final constant C.

Applying Lemma 3 with

$$N = L, \qquad D_r = D, \qquad D_{r+1} = \frac{(r+1)D}{M}, \qquad \delta = \Delta, \qquad W = 2,$$

and absorbing the harmless powers of $W = 2$ into the absolute constant gives the stated estimate. □

3 Proof of the main theorem

Proof of Theorem 1. Fix

$$\beta = \frac{1}{10}.$$

This choice is smaller than 0.465, so Lemma 2 applies. Put

$$X = \log k, \qquad Y = \log X.$$

Choose a positive constant c sufficiently small so that

$$\frac{\beta}{15c} > 20, \qquad \frac{1}{20c} > 20. \tag{C}$$

We shall prove that, for all sufficiently large k, no k-admissible integer n can satisfy

$$2k < n \leq \exp\left(c\frac{X^2}{Y} \right). \tag{2.1}$$

Assume, for contradiction, that such a k-admissible n exists for arbitrarily large k. By Lemma 1,

$$n \geq c_1 \frac{k^2}{X}. \tag{2.2}$$

Let

$$H = \lfloor k^{1-\beta} \rfloor, \qquad \Delta = 2k^{-\beta}.$$

For sufficiently large k, we have $H < k$, and hence

$$(k, k+H] \subset (k, 2k).$$

Let $p \in (k, k+H]$ be prime, and let r_p be the least nonnegative residue of n modulo p. Since n is k-admissible,

$$r_p \notin \{1, 2, \dots, k\}.$$

Thus either $r_p = 0$, or $k+1 \le r_p \le p-1$. If $r_p = 0$, then

$$\left\| \frac{n}{p} \right\| = 0.$$

If $r_p > k$, then

$$\left\| \frac{n}{p} \right\| \le 1 - \frac{r_p}{p} \le 1 - \frac{k+1}{p} < \frac{p-k}{p} \le \frac{H}{k}.$$

Therefore, for every prime $p \in (k, k+H]$,

$$\left\| \frac{n}{p} \right\| \le \frac{H}{k} < \Delta.$$

Since $H = \lfloor k^{1-\beta} \rfloor$, Lemma 2, applied with $x = k$, gives

$$\#\{p \text{ prime} : k < p \le k+H\} \gg \frac{H}{X}.$$

Hence

$$\mathcal{N} := \left| \left\{ u \in (k, k+H] \cap \mathbb{Z} : \left\| \frac{n}{u} \right\| < \Delta \right\} \right| \gg \frac{H}{X}. \tag{2.3}$$

We shall now contradict this lower bound by applying Lemma 4.

For $j \ge 2$, define

$$Q_j = \frac{nj!}{k^{j+1}}.$$

Let r be the least integer $j \ge 2$ for which

$$Q_j \le k^{-\beta}. \tag{2.4}$$

Such an r exists for all sufficiently large k. Indeed, if

$$R = \left\lceil \frac{4cX}{Y} \right\rceil + 10,$$

then, using (2.1) and $\log(R!) \le R \log R$, we get

$$\log Q_R \le c\frac{X^2}{Y} + R \log R - (R+1)X \le -2c\frac{X^2}{Y}$$

for sufficiently large k. In particular $Q_R \le k^{-\beta}$, once k is large. Thus

$$2 \le r \le R \le 5c\frac{X}{Y} \tag{2.5}$$

for sufficiently large k.

Since $H/k \le k^{-\beta}$, it follows from (2.5) that

$$\frac{(r+1)H}{k} = o(1).$$

In particular,

$$\frac{(r+1)H}{k} \leq \log 2$$

for all sufficiently large k. Hence Lemma 4 may be applied with

$$M = k, \qquad L = H, \qquad A = n, \qquad D = Q_r, \qquad \Delta = 2k^{-\beta}.$$

Set

$$a_r = \frac{2r-1}{3r-2}$$

and choose

$$\lambda = \left(\frac{\Delta^{a_r}}{Q_r}\right)^{1/r}. \tag{2.6}$$

Since $a_r < 1$, $\Delta = 2k^{-\beta}$, and $Q_r \leq k^{-\beta}$, we have $\lambda \geq 1$ for all sufficiently large k. Lemma 4 gives

$$\mathcal{N} \leq CH\left[(Q_r\lambda^r)^{1/(2r-1)} + \left(\frac{\Delta}{Q_r\lambda^r}\right)^{1/(r-1)} + \left(\frac{(r+1)\lambda}{k}\right)^{1/(2r)}\right] + 2r\lambda. \tag{2.7}$$

By the definition of λ, the first two terms in the square brackets are both equal to

$$\Delta^{1/(3r-2)}.$$

Using (2.5) and (C), we get

$$\Delta^{1/(3r-2)} \ll \exp\left(-\frac{\beta X}{3r}\right) \leq \exp\left(-\frac{\beta Y}{15c}\right) = X^{-\beta/(15c)} \ll X^{-20}. \tag{2.8}$$

It remains to estimate the third term in (2.7) and the final term $2r\lambda$.

First suppose $r = 2$. Then $Q_2 = 2n/k^3$, and (2.2) gives

$$Q_2 \gg \frac{1}{kX}.$$

Since $a_2 = 3/4$, it follows from (2.6) that

$$\lambda \ll \left(k^{-3\beta/4} \cdot kX\right)^{1/2} = X^{1/2}k^{1/2-3\beta/8}. \tag{2.9}$$

Therefore

$$\left(\frac{3\lambda}{k}\right)^{1/4} \ll \left(X^{1/2}k^{-1/2-3\beta/8}\right)^{1/4} \ll k^{-1/10} \ll X^{-20}, \tag{2.10}$$

and also

$$r\lambda \ll X^{1/2}k^{1/2-3\beta/8} = o\left(\frac{H}{X^2}\right), \tag{2.11}$$

because $H \asymp k^{1-\beta}$ and

$$1 - \beta - \left(\frac{1}{2} - \frac{3\beta}{8}\right) = \frac{1}{2} - \frac{5\beta}{8} > 0.$$

Now suppose $r \geq 3$. By the minimality of r,

$$Q_{r-1} = \frac{n(r-1)!}{k^r} > k^{-\beta}.$$

Since $Q_r = (r/k)Q_{r-1}$, we have
$$Q_r > rk^{-\beta-1}.$$
Also
$$1 - a_r = \frac{r-1}{3r-2} < \frac{1}{3}.$$
Hence (2.6) gives
$$\lambda^r = \frac{\Delta^{a_r}}{Q_r} \ll \frac{k^{-\beta a_r}}{rk^{-\beta-1}} \le k^{1+\beta/3},$$
and therefore
$$\lambda \ll k^{(1+\beta/3)/r}. \tag{2.12}$$
For $r \ge 3$ and $\beta = 1/10$, the exponent $(1+\beta/3)/r$ is at most $31/90$. Since $r \le 5cX/Y$, we have $r+1 \le k^{1/20}$ for sufficiently large k. Thus, by (2.12),
$$\frac{(r+1)\lambda}{k} \ll k^{-1+1/20+31/90} \le k^{-1/2}$$
for all sufficiently large k. Consequently,
$$\left(\frac{(r+1)\lambda}{k}\right)^{1/(2r)} \ll k^{-1/(4r)} \le \exp\left(-\frac{Y}{20c}\right) = X^{-1/(20c)} \ll X^{-20}, \tag{2.13}$$
where we used (2.5) and (C).

Finally, still assuming $r \ge 3$, (2.12) gives
$$r\lambda \ll rk^{(1+\beta/3)/r} \le rk^{31/90}.$$
Since $H \asymp k^{9/10}$ and $r \ll X/Y$, this implies
$$r\lambda = o\left(\frac{H}{X^2}\right). \tag{2.14}$$

Combining (2.7)–(2.14), we obtain
$$\mathcal{N} \ll \frac{H}{X^{20}} + o\left(\frac{H}{X^2}\right) \ll \frac{H}{X^2}. \tag{2.15}$$
But (2.3) gives
$$\mathcal{N} \gg \frac{H}{X}.$$
Equivalently, there exists an absolute constant $c_* > 0$ such that
$$\mathcal{N} \ge c_* \frac{H}{X}$$
for all sufficiently large k. On the other hand, (2.15) yields
$$\mathcal{N} \le C_* \frac{H}{X^2}$$
for some absolute constant $C_* > 0$. Hence
$$c_* \le \frac{C_*}{X},$$
which is impossible for sufficiently large k, since $X = \log k \to \infty$. $\square$

Remark 1. *The proof uses only primes p in the short interval*

$$(k, k + k^{1-\beta}].$$

For each such prime, admissibility forces n/p to be very close to an integer. Konyagin's theorem says that this can happen for at most $O(H/(\log k)^2)$ integers $u \in (k, k+H]$ when

$$n \leq \exp\left(c \frac{(\log k)^2}{\log \log k} \right),$$

whereas the Baker–Harman theorem supplies $\gg H/\log k$ primes in the same interval. The remaining logarithmic loss comes from the fact that we only get $H/\log k$ prime-produced points, not H points.

References

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